PROJECT REPORT

TRAFFIC ANALYTICS & TRAVEL TIME PREDICTION

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Computer Applications (BCA)

Jaypee University

Area: Data Science

Internship Duration: 2 Months

Submitted By

Rishabh Gaur

Ankit Yadav

Himanshi Yadav

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Jaypee University for providing the opportunity to work on this project.

I am thankful to my project guide and faculty members for their valuable guidance, suggestions, and support throughout the development of this project.

I would also like to thank my friends and family for their continuous encouragement and support during the completion of this project.

ABSTRACT

Traffic congestion is a major challenge in modern transportation. Traffic conditions change according to time, traffic volume, vehicle speed, weather, and distance.

The Traffic Analytics & Travel Time Prediction project focuses on analyzing traffic-related data and developing a machine-learning-based approach for predicting travel time.

The project includes data loading, preprocessing, exploratory data analysis, visualization, feature preparation, model development, prediction, and evaluation using Python.

The current implementation uses a small sample dataset and demonstrates how traffic information can be converted into useful analytical and predictive results.

1. INTRODUCTION

Traffic management is an important application area of data science. Traffic conditions can change depending on time, traffic volume, road conditions, weather, and other factors.

Data-driven approaches can identify historical patterns and use them to estimate travel time. This project demonstrates a complete workflow from traffic data analysis to machine-learning-based travel-time prediction.

2. PROBLEM STATEMENT

Accurately predicting travel time is difficult because traffic conditions change continuously. Major challenges include variable traffic volume, peak-hour congestion, changes in vehicle speed, weather effects, and unexpected traffic incidents.

The project aims to use traffic-related features to analyze these patterns and predict travel time.

3. OBJECTIVES

To analyze historical traffic data.

To understand traffic patterns by hour.

To study relationships between traffic volume, speed, and travel time.

To preprocess traffic data for machine learning.

To develop a travel-time prediction model.

To evaluate model performance using regression metrics.

To visualize traffic and prediction results.

4. SCOPE OF THE PROJECT

The project covers traffic data analysis, visualization, preprocessing, feature engineering, machine learning, travel-time prediction, and model evaluation.

The project can be extended in the future with real-time GPS data, traffic APIs, live weather information, incident data, and web dashboards.

5. DATASET DESCRIPTION

The implemented project uses a sample dataset containing 32 observations and 8 columns.

Columns: Date, Hour, Day, Traffic_Volume, Average_Speed, Weather, Distance_km, Travel_Time_min.

Target variable: Travel_Time_min.

Model input features: Hour, Traffic_Volume, Average_Speed, Weather, Distance_km.

The current dataset is a small academic sample, so the results should be treated as a demonstration of the workflow.

6. TECHNOLOGIES USED

Python: Programming and data analysis.

Pandas: Data loading and manipulation.

NumPy: Numerical operations.

Matplotlib / Seaborn: Data visualization.

Scikit-learn: Machine learning and evaluation.

Visual Studio Code / Jupyter Notebook: Development.

Git / GitHub: Version control and project submission.

7. SYSTEM REQUIREMENTS

Hardware: Computer with at least 4 GB RAM, adequate storage, and internet access.

Software: Windows/Linux/macOS, Python 3.x, required Python libraries, and a code editor or Jupyter Notebook.

8. DATA PREPROCESSING

The data is loaded using Pandas and prepared for analysis.

The Weather feature is encoded numerically, with Clear represented as 0 and Rain as 1.

The model uses Hour, Traffic_Volume, Average_Speed, Weather, and Distance_km as input features and Travel_Time_min as the target.

The data is divided into training and testing sets using an 80:20 split.

9. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis is used to understand traffic behaviour and identify patterns.

The project analyzes traffic volume by hour, travel time by hour, traffic volume versus travel time, and average speed versus travel time.

These visualizations help explain how traffic conditions are associated with journey duration.

10. FEATURE ENGINEERING

Useful features for traffic prediction can include hour, day of week, month, weekend indicator, peak-hour indicator, traffic volume, speed, weather, and distance.

The implemented model uses Hour, Traffic_Volume, Average_Speed, Weather, and Distance_km.

11. MACHINE LEARNING MODEL

The implemented project uses Random Forest Regression for travel-time prediction.

Random Forest combines multiple decision trees and can capture nonlinear relationships between input variables.

The training set is used to learn patterns and the testing set is used to evaluate prediction performance on unseen observations.

12. MODEL EVALUATION

Mean Absolute Error (MAE): 1.08 minutes.

Mean Squared Error (MSE): 1.56.

R^2 Score: 0.98.

These results are from the current 32-observation sample and should not be interpreted as production-level performance.

13. RESULTS AND FINDINGS

Average Traffic Volume: 1754.69.

Average Speed: 27.69 km/h.

Average Travel Time: 31.09 minutes.

Predicted Travel Time in the example: 44.46 minutes.

The project demonstrates that traffic volume, speed, hour, weather, and distance can be used as features for estimating travel time.

14. ADVANTAGES

The project provides a structured traffic-analysis workflow.

It helps identify traffic patterns and peak periods.

It demonstrates machine-learning-based travel-time prediction.

It can be extended to real-time transportation applications.

15. LIMITATIONS

The current dataset contains only 32 observations.

Unexpected incidents may not be represented.

Model performance can vary for different roads and locations.

Real-time prediction requires continuously updated traffic data.

16. FUTURE SCOPE

Integration with real-time traffic APIs.

GPS and live vehicle-location data.

Live weather and road-condition information.

Real-time congestion alerts.

Web-based dashboards and mobile applications.

Advanced machine-learning and deep-learning models.

Cloud deployment for real-time prediction.

17. CONCLUSION

The Traffic Analytics & Travel Time Prediction project demonstrates the practical use of data science and machine learning in transportation.

The workflow covers data preprocessing, exploratory analysis, visualization, feature preparation, Random Forest Regression, prediction, and evaluation.

The project provides a foundation for a larger real-time traffic prediction system when more extensive and continuously updated data is available.

18. REFERENCES

Python Documentation.

Pandas Documentation.

NumPy Documentation.

Scikit-learn Documentation.

Matplotlib Documentation.

Seaborn Documentation.

Academic literature related to traffic forecasting and travel-time prediction.